Quiz 1

Solution

(i) $f+g = \sqrt{x} + x^2 - 2x - 3 \Rightarrow$ Domain $\mathbb{R}^+$

(ii) $f-g = \sqrt{x} - x^2 + 2x + 3 \Rightarrow$ Domain $\mathbb{R}^+$

(iii) $f \times g = (\sqrt{x})(x^2 - 2x - 3) \Rightarrow$ Domain $\mathbb{R}^+$

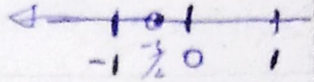
(iv). $f/g = \sqrt{x}/(x^2 - 2x - 3) \Rightarrow$ Domain $\mathbb{R}^+ - \{3\}$

Q2 (a) $\frac{3x+1}{x-2} < 1 \Rightarrow 3x+1 < x-2$

$3x - x < -3$

$2x < -3$

$x < -3/2$

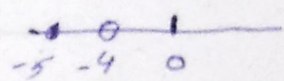


(b) $x^2 + 9x + 20 \leq 0$

$x^2 + 5x + 4x + 20 \leq 0$

$x(x+5) + 4(x+5) \leq 0$

$(x+5)(x+4) \leq 0$



$x \in (-\infty, -5) \Rightarrow - \times - = + \geq 0$

$x \in [-5, -4] \Rightarrow + \times - = - \leq 0$

$x \in (-4, \infty) \Rightarrow + \times + = + \geq 0$

Hence $[-5, -4]$ is the required int

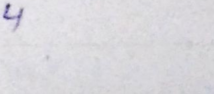
(c) $|3x+1| < 4$

$-4 < 3x+1 < 4$

$3x+1 > -4$ $\quad$ $3x+1 < 4$

$\boxed{x > -1}$ $\quad$ $\boxed{x < 1}$

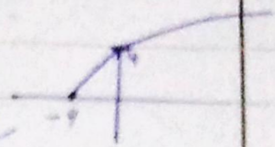
$-1 < x < 1$



(Q3) Given Relation is function
for each year there is a unique output

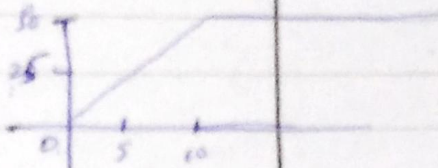
• Since the relation is not one-one
Inverse relation is not a function.

(Q4) a) $f = 2\sqrt{x+4}$ Domain $x \geq -4$
Range $y \geq 0$ using graph



$y = 2\sqrt{x+4}$

(b) $C(n) = \begin{cases} 5n & 0 < n < 10 \\ 50 & n \geq 10 \end{cases}$



Domain $n \in (0, \infty]$ Range $(0, 50]$
$0 < n < \infty$ $0 < C(n) \leq 50$

Q5 $\lim_{x \to 5} \frac{x^2 - 25}{x - 5}$

$5-2\delta$	$5-\delta$	5	$5+\delta$	$5+2\delta$
9.998	9.999	X	10.001	10.002

$\delta = 0.01$

Q6 $\lim_{x \to 5} \frac{1/x - 1/5}{x - 5}$

$\frac{5-x}{5x} / x-5 = -\frac{1}{5x}$

$= \frac{-1}{25}$

$\lim_{x \to 0} \frac{\sqrt{25-x} - 5}{x} \times \frac{\sqrt{25-x} + 5}{\sqrt{25-x} + 5}$

$\lim_{x \to 0} \frac{(\sqrt{25-x})^2 - 25}{x \times (\sqrt{25-x} + 5)} = \frac{-x}{x \times (\sqrt{25-x}+5)}$

$\lim_{x \to 0} \frac{-1}{\sqrt{25-x} + 5} = \frac{-1}{10}$ Ans

(Q7) (i) $\lim_{x \to 0}$ exist $f(0)$ is not defined
∴ It Removable discontinuity